Electric Air Shipping

JELLYFISH

A MONTHLY UPDATE

NO. 003 — OCTOBER 2026



EAS is decarbonising global freight with ultra-heavy-lift cargo airships.

IN THIS ISSUE

01 The Royal Aeronautical Society

02 ARENA Launchpad application submitted

03 We're not leaving (in fact, we're growing)

Explainer The square-cube law – where shape meets size

04 Toby's Thoughts Brian's Thoughts

05 Airship news: Flying Whales comparison

01 RAES ADELAIDE – 8 OCTOBER

The Royal Aeronautical Society



The RAeS Adelaide audience after the talk, 8 October. Photo: James Blagg.

On 8 October Toby and Brian presented "Decarbonising Global Freight with Cargo Airships" to the Royal Aeronautical Society, Adelaide Chapter. Toby ran the EAS fleet calculator live during the talk (calc.electricairshipping.com).

It was a great night. It was a great response. We got lots of questions. We were told it was inspirational. We were introduced as 'like meeting the Wright brothers'. (!) Put it this way: In a room full of Adelaide's best and brightest aero-engineers, there was not a single question we couldn't answer to a level that felt satisfactory to the whole room. And afterwards, people were really happy to endorse us, chat over dinner, and the feedback was amazing. For EAS, and us personally, it was quite a big deal.

> "Tonight was a rare moment in life where I learnt of new concepts in sustainable international airfreight, based on a whole new approach to design of super sized airships. The concepts are akin to first exposure to a jet engine, or a multi stage rocket."
>
> JAMES BLAGG CSC, CHAIR OF THE RAES ADELAIDE CHAPTER

02 ARENA LAUNCHPAD

ARENA Launchpad application submitted

On 28 October EAS submitted an application to the Australian Renewable Energy Agency's Launchpad Startup Funding Round, for a 2027 engineering validation program. Results are expected in mid-December.

It was a huge amount of work. We had to get quotes. We had to create and scope the project. We had to put together all the documentation. We had to find people who would agree to work with us. And now we're basically just praying that our thing is successful.

We think it's the biggest upside project that anyone would have submitted, and it's very aligned to ARENA – to decarbonising transport and creating new Australian capability. So we hope that we're in with a good chance.

LaunchPad
POWERED BY ARENA

> *...to accelerate the development of early-stage Australian companies and innovators...*
>
> ARENA LAUNCHPAD PROGRAM GUIDELINES, SEPTEMBER 2026

TRL 3 → 4

STATIC + DYNAMIC ENGINEERING VALIDATION

STATIC

Can it be built at scale?

Tessellation · FEA · test models · load testing

STRUCTURAL WEIGHT

⇄

DYNAMIC

Can it cruise efficiently?

CFD · wind tunnel

DRAG COEFFICIENT (CD) · BLI POWER SAVING (S)

↓

Software ⇄ Hardware → **Data** → **Fleet calculator update**

↓

Energy · emissions · unit economics · fleet requirements and projections

NEXT PRESENTATION – SUSTAINABLE ENGINEERS GROUP (SENG) ENGINEERS AUSTRALIA (EA)

Sustainable long-distance freight by electric airships

Thursday 26 November 2026, 6.00–7.00 pm Adelaide time
Level 11, 108 King William Street, Adelaide, or watch online
Open to the public. Register at engineersaustralia.org.au

03 THE OFFICE

We're not F***ing leaving!

EAS office lives to fight another day



We love the office, and after sadly accepting last month that we'd be moving out, a few things changed our mind, and we've decided to stay on with a month-by-month type agreement.

Partly it's because we've got some interns coming. Partly it's because we were so busy with our ARENA application and with other things happening over the next few months. Also to keep John company next door, and to have our own room to work from. We'll reassess at the start of 2027.

(IN FACT, WE'RE GROWING)

Joyce, a voluntary mechanical engineering intern, joins EAS for 12 weeks from 1 December (via CareerDC).

We got a master agreement back today with the Adelaide University business college (WIL). So in addition to Joyce as a voluntary mechanical engineer, we're working towards interns from business and engineering also.

The whole EAS project is such a research white space that it overlaps with almost every college and almost every discipline in some regard, because our configuration hasn't been studied – from engineering to architecture to business to all sorts of materials science modelling, like climate and economic modelling, et cetera.

If anybody sees this newsletter and is interested, or any university students who would like to work on our projects – please get in touch. You may be some of the first external researchers, workers or helpers of the new aerostatics field and industry.

contact@electricairshipping.com

EXPLAINER 03 – SIZE, SCALE, AND THE SQUARE-CUBE LAW 03 OF 12

The square-cube law – where shape meets size

(Almost anything could be an airship if it was big enough, and full of hot air.)

Last edition, we took every shape in the universe, put it inside of a ball and spun it around. In doing so, we got VS and VE. In this edition, SIZE appears – and with size (AKA scale), we get real volume, in real units.

So today we add a measuring stick. In volume land however, the measuring stick is a measuring sphere - because for any given size, no shape can hold more volume than a sphere. So a sphere IS volume. And we're very interested in volume, because volume gives us buoyant lift.

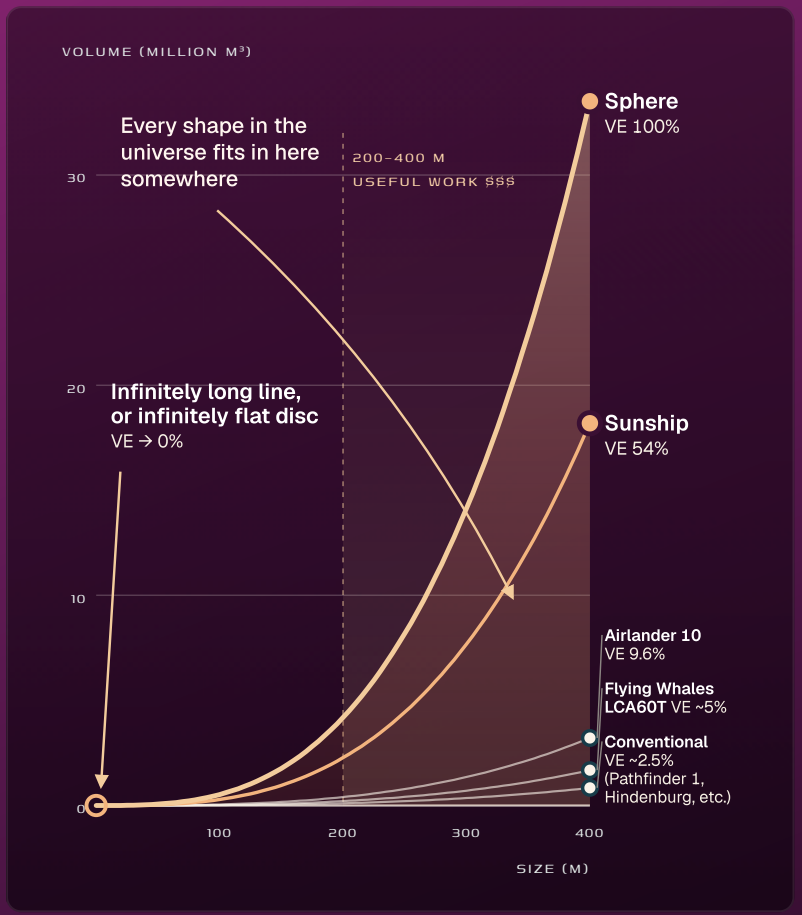
Things are big or small relative to other things. As humans, things are big or small relative to us. From a different frame of reference, something very large may appear tiny, and vice versa.

But there are limits. Our limit length is 400 metres, the size of the biggest container ships. But not a 400 metre measuring stick, a 400 metre measuring sphere. So the maximum volume for buoyant lift is around 33 million cubic metres, the volume of a 400 metre sphere.

Every shape in the universe, if placed inside this giant maximum sphere would have some volume – but not every shape would have enough volume to do useful work. This is the point where having a really good VE percentage, like the Sunship, means useful work can be done without having to build an airship that is insanely, stupidly, impossibly big. Just *really* big, and no bigger. This is important.

So shape is the thing through which the square-cube law is expressed, and no shape expresses the square-cube law more efficiently than the sphere. In other words, if you get the shape correct when it's very small, the difference becomes really significant when that shape becomes gigantic.

Think of anything you can see - made to the size of a skyscraper. Some things scale better than others - and very few things are designed to be gigantic. The Sunship is, and as size increases, basically every single performance measure improves. Aerodynamics? Structural stability? Landing, wind, weather? These concerns can be addressed later - only after lifting enough payload to provide truly useful work, for humans here on Earth.





TOBY'S THOUGHTS

TOBY BRANDENBURG

Short one this month. ARENA grant was intense - and we are 'the men in the ARENA'. Thoughts about the 'persona' of success and being pulled towards work as opposed to pushing. Tiredness with a full soul, as opposed to exhaustion and burnout. Blimp daddy loading.

More people should email or message me. Get in touch peeps!

PEACE – TOBY B

BRIAN'S THOUGHTS

BRIAN BRANDENBURG

Welcome to Jellyfish number 3. The 31st of October is here – Halloween for so many, to such extremes as we integrate ever more with the United States. A month of significant milestones and achievements for Electric Air Shipping.

October the 8th was very significant. The regular meeting of the Royal Aeronautical Society went very, very well. A big shout-out to James Blagg, Chair of the Adelaide branch. The meeting was in a large and impressive lecture theatre with amazingly large, clear, bright screens. After the usual technical issues with the screen, internet access and so on, which Toby, James and Charles Giacco dealt with so quickly, I was gifted with a lifelong memorable moment when James commenced his introduction by comparing Toby and me to the Wright brothers. I'll never forget that.

It was amazing to see HTA-trained aeronautical engineers and their oncoming uni students expressing such enthusiasm for our breakthrough in LTA transport. James informed us that airships were involved in the early years of the Royal Aeronautical Society, which was founded in the 1860s, 40 years before the Wright brothers' first flight in 1903.

Toby and I are considering affiliate membership of the Royal Aeronautical Society, and are investigating the publication of an article in its journal to inform all of its members – 25,000 of them now – of both the details of our Sunship, and the size of the project and of the market it will generate, and the large numbers of engineering graduates and workers it will require.

The decarbonisation focus of the project is so important – and also the reduction of maritime pollution, which gets really extreme at the world's choke points, and the sulphur dioxide emissions that ships pour out into the ports of the world.

In the long term, the replacing of virtually all emissions-emitting planes and maritime ships will be a possibility, along with an entire industry being established for fresh/rainwater collection and transport, to solve the planet's massive water shortage and rapid aquifer depletion. "Water Bankruptcy".

Each month from now on, I plan to write in Jellyfish an article on the achievements, milestones and new contacts that we've made, and the effect they will have on increasing our valuation in steady increments every month. I'll be using the comparison valuation method – comparing our project with similar-sized projects.

And while we're here – similar-sized projects – literally on page 7 of this month's Jellyfish, we talk about Flying Whales, who raised another 90 million euro for a total of about 400 million euro. In some ways, Flying Whales aren't much past TRL 3–4 - the level we aim to achieve in 2027, especially if we are successful in our ARENA application. Our Sunship design, however, has projected freight performance orders of magnitude above that of the LCA60T and effectively any other conventional-based airship design.

Our valuation will be steadily increasing from TRL 3 to TRL 4 over the coming 12 months or so. I feel that the valuation doesn't jump up – it will go up in steps each month as we achieve tasks.

In future, I want to really investigate similar companies with similarly large total addressable markets, e.g. Uber, Tesla and SpaceX, to look at the valuations they achieved at pre-seed and seed, and at TRL 3, 4 and 5, and compare with them, because our comparison is so strong.

CHEERS – BRIAN B

AIRSHIP NEWS: FLYING WHALES COMPARISON

ASSOCIATED PRESS, 23 OCTOBER 2026

US designates helium as a critical material, citing its importance for semiconductors

The US Department of Energy has added helium to its list of critical materials, the 20th on the list. It cites helium's role in chip manufacturing and a real risk to global supply: Qatar's exports have been cut sharply by the closure of the Strait of Hormuz, and Russia imposed temporary export controls in April.

> *Which is very bad news if your business model is filling up gigantic bags of helium to lift things around.*

FRENCH TECH JOURNAL, 19 OCTOBER 2026

Flying Whales' €90M led 15 deals for €215M

French cargo-airship developer Flying Whales closed a €90 million round on 12 October, led by the shipping line CMA CGM, taking its total funding to about €400 million. The money goes to its LCA60T, a helium airship designed to carry up to 60 tonnes, and a factory near Bordeaux. It plans to start assembly in early 2028, with a first test flight late that year.

> *On a completely unrelated note (wink) – it's good that there's money out there for airship freight. We basically can't wait until the world realises that you can actually do more with less, and we can lift 10… 50… even 100 or 200 times that amount with hot air, with our Sunship configuration.*

SAME LENGTH: 200 M

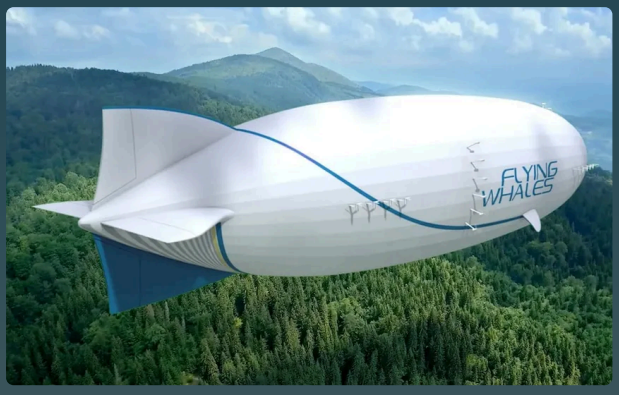


Image: Flying Whales

VS



The Sunship, EAS render

Flying Whales LCA60T · helium · VS 16 · VE 4.8–6.0% · 200,000–250,000 m³ · gross lift ~235 t · payload fraction ~25% · **60 t payload**

EAS Sunship · hot air, 500 °C · VS 1.6 · VE 54.3% · about 2.3 million m³ · gross lift ~1,750 t · payload fraction ~58% · **~1,000 t payload**

> *At 200 metres long – the same length – the Sunship would be in the order of 1,000 tonnes, with variable buoyancy from hot air, and no critical material helium involved. That's ridiculous.*

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FEED ME TO YOUR AI → electricairshipping.com/llms-full.txt